

ENVS193DS Final

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Table of contents

GitHub repo	2
Set up	2
Problem 1. Research writing	3
a. Transparent statistical methods	3
b. Suggestions for rewriting	3
c. Further analysis	3
Problem 2. Data analysis	4
a. Clean and wrangle	4
b. Response variable	5
c. Purpose of study	5
d. Table of models	5
e. Run the models	6
f. Select the best model	6
g. Check the diagnostics	6
h. Visualize the model predictions	7
i. Write the caption for your figure	10
j. Calculate model predictions	10
k. Interpret your results	11
Problem 3. Affective and exploratory visualizations	11
a. Comparing visualizations	11
b. Designing an analysis	12
c. Check any assumptions and run your analysis	12
Think about your data	12
Write your model	12
Look at the diagnostics	13

Look at the model coefficients	14
Look at the model predictions	15
d. Create a visualization	16
Visualize model predictions	16
Describe the model	18
e. Write a caption	19
f. Write about your results	19
g. Sharing your affective visualization	19

GitHub repo

Access my repo [here](#).

Set up

```
# read in packages
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(DHARMA) # for GLM diagnostics
library(ggeffects) # for ggpredict
library(flextable) # for flextable
library(gtsummary) # for GLM
library(MuMin) # for AIC
library(gt) # for gt summary table

# read in data
nests <- read_csv(here("data", "nest_data_final.csv"))
# read in personal data to object from csv
my_data <- read_csv(here("data", "envs193ds-personal-data.csv"))

# setting up personal data for problem 3

# cleaned column names to snake case
my_data <- my_data |>
  clean_names()

# created new column for binary variable: 1 if workout was a run, 0 if not
my_data <- my_data |>
```

```
mutate(is_run = case_when(  
  workout_type == "Run" ~ 1, # run = 1  
  .default = 0))           # all other workout types = 0
```

Problem 1. Research writing

a. Transparent statistical methods

Part 1: In Part 1, the co-worker used a binomial generalized linear model (logistic regression). The response variable is a binary outcome reflecting whether the American Avocet uses the microhabitat (1) or does not (0), and the predictor is a continuous variable: distance to water edge (meters).

Part 2: In Part 2, the co-worker used a chi-square test of independence to ask whether a relationship exists between two categorical variables: bird species (American Avocet, Black-necked Stilt, Forster's Tern) and habitat type (managed pond, non-tidal marsh, salt pond, tidal marsh, tidal mudflat). The test compares observed counts to expected counts to determine if the two variables are associated.

b. Suggestions for rewriting

Part 1: American Avocets were more likely to use the microhabitats that were closer to the water's edge, and this relationship was strong enough that it is unlikely to be the result of random chance alone.

Part 2: The three shorebird species did not use the five habitat types in equal proportions: each species showed a distinct pattern of habitat preference, and these differences across species are unlikely to reflect chance.

c. Further analysis

Tidal inundation frequency is a continuous variable measured as the proportion of time a microhabitat is flooded per tidal cycle, recorded using tidal monitoring stations. Microhabitats that are too rarely or too frequently inundated may not provide the shallow water depth conditions that Avocets need for successful foraging.

Problem 2. Data analysis

a. Clean and wrangle

```
# created new object from nests
nests_clean <- nests |>
# filtered to include ant species: C. mimosae, C. sjostedti, and C. nigriceps
filter(ant_species %in% c("RRB", "AB", "BBR")) |>
# created new column with scientific name
mutate(species_name = recode_values(ant_species,
                                   "RRB" ~ "Crematogaster mimosae",
                                   "AB" ~ "Crematogaster sjostedti",
                                   "BBR" ~ "Crematogaster nigriceps")) |>

# reordered ant species
mutate(species_name = as_factor(species_name),
       species_name = fct_relevel(species_name,
                                   "Crematogaster sjostedti",
                                   "Crematogaster mimosae",
                                   "Crematogaster nigriceps")) |>

# selected columns of interest
select(height_cm, case_control, species_name)

# checked structure
str(nests_clean)
```

```
tibble [270 x 3] (S3: tbl_df/tbl/data.frame)
 $ height_cm    : num [1:270] 392 310 513 154 324 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ species_name: Factor w/ 3 levels "Crematogaster sjostedti",...: 2 2 2 2 1 2 3 2 1 3 ...
```

```
# displayed ten random rows
slice_sample(nests_clean, # cleaned data frame to display
             n = 10)      # showed ten rows
```

```
# A tibble: 10 x 3
  height_cm case_control species_name
    <dbl>         <dbl>    <fct>
1     247             1 Crematogaster mimosae
2     446             0 Crematogaster mimosae
3     154             0 Crematogaster mimosae
```

4	195	1 Crematogaster mimosae
5	159	0 Crematogaster mimosae
6	324.	0 Crematogaster sjostedti
7	332.	0 Crematogaster mimosae
8	201	1 Crematogaster mimosae
9	130	0 Crematogaster mimosae
10	166.	0 Crematogaster mimosae

b. Response variable

The response variable `case_control` is binary and describes whether a tree in the survey contained a nest (1) or was an “available” tree for birds to nest in (0).

c. Purpose of study

- **Tree height** is a continuous predictor variable measured in centimeters. As tree height increases, the probability of nest occurrence in the tree would increase because taller trees may provide greater protection from predators on the ground that threaten nesting birds (Results, paragraph 2). Taller trees may also offer more canopy structure for nest placement (Discussion, paragraph 5).
- **Acacia ant species** is a categorical predictor variable with three levels: *C. sjostedti*, *C. mimosae*, and *C. nigriceps*. Ant species could predict nest occurrence because different ant species vary in their aggressiveness toward mammals on the ground that threaten nesting birds, thereby altering the level of protection available to nesting birds (Introduction, paragraph 2). While *C. mimosae* and *C. nigriceps* aggressively defend trees against mammals, insect herbivores, and lethal herbivory by elephants, *C. sjostedti* only provides minimal protection against herbivory (Introduction, paragraph 2). Therefore, nesting birds may preferentially select trees occupied by *C. mimosae* and *C. nigriceps*, which offer greater protection, reflecting a mutualistic relationship between these ant species and birds.

d. Table of models

Model number	Tree Height (cm)	Ant Species	Model description
1	X	X	all predictors, no interaction
2	X	X	all predictors, interaction term

e. Run the models

```
# model 1: no interaction
# formula: case_control as a function of tree height plus ant species
model1 <- glm(case_control ~ height_cm + species_name,
  # data: nests_clean data frame
  data = nests_clean,
  # binomial error distribution for binary response
  family = binomial)

# model 2: interaction term (tree height times ant species)
# formula: case_control as a function of tree height times ant species
model2 <- glm(case_control ~ height_cm * species_name,
  # data: nests_clean data frame
  data = nests_clean,
  # binomial error distribution for binary response
  family = binomial)
```

f. Select the best model

```
# calculated AIC
AICc (model1,
  model2) |>
# arranged by decreasing AIC values
arrange (AICc)
```

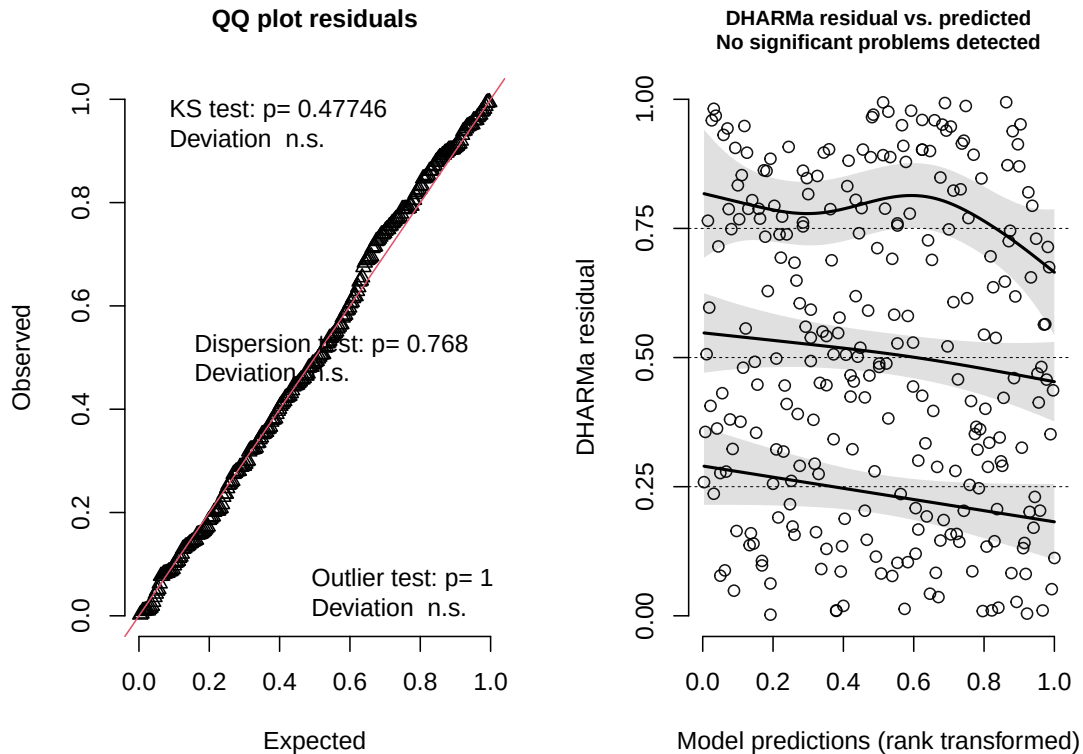
	df	AICc
model2	6	238.1236
model1	4	258.8940

Conclusion: The best model as determined by Akaike's Information Criterion (AIC) is the model predicting probability of nest occurrence from tree height and ant species with an interaction term (tree height $\times$ ant species); this is model 2 in my code.

g. Check the diagnostics

```
# simulated residuals from DHARMA for model 2
plot(simulateResiduals(model2))
```

DHARMA residual



Information from QQ plot of residuals: The residuals appear uniformly distributed because the points closely follow the reference line matching the quantiles of simulated residuals with quantiles of uniform distribution. The residuals do not appear more dispersed than expected from a uniform distribution, and there does not appear to be any outliers in the data set that could be influencing model coefficients or predictions.

Information from DHARMA residual vs predicted: Residuals appear randomly distributed across the range of predicted values, with no clear pattern of concern.

h. Visualize the model predictions

```
# generated model predictions across range of height values for each species
preds <- ggpredict(
  model2, # model object 2
  terms = c("height_cm [all]", "species_name")) |> # predictors
# renamed columns
rename(height_cm = x,
        species_name = group)
```

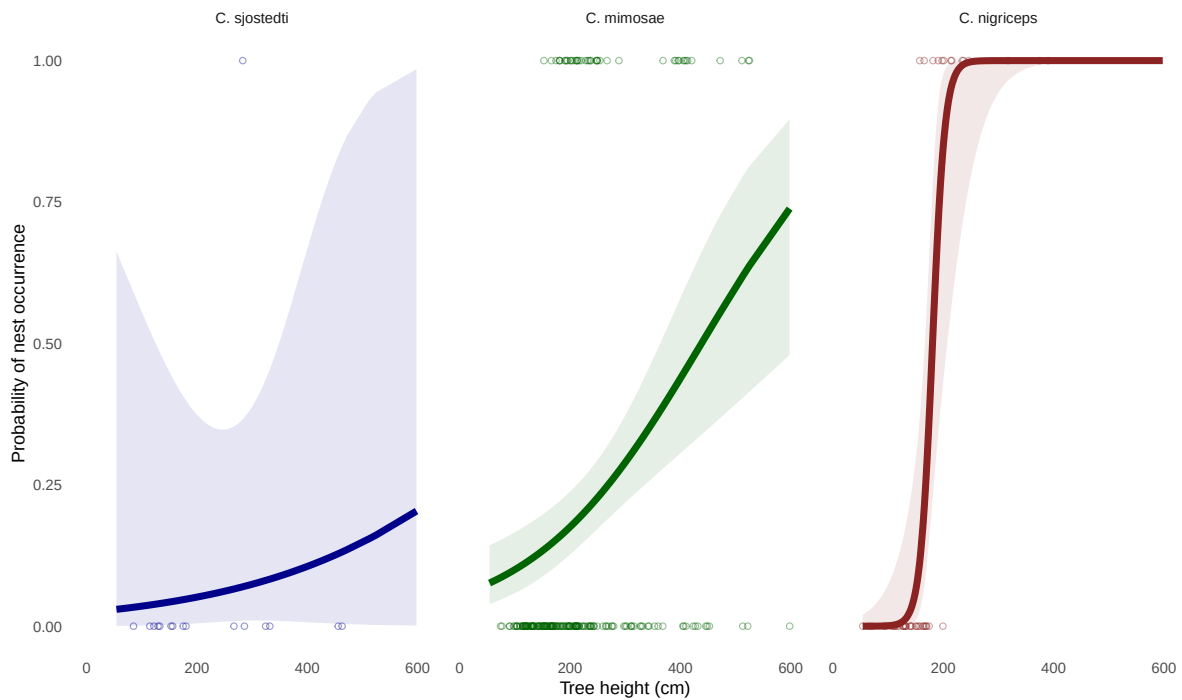
```
# base layer: ggplot
ggplot(data = nests_clean, # data frame: nests_clean
       mapping = aes(x = height_cm, # x-axis: height (cm)
                     y = case_control)) + # y-axis: presence of nest (0/1)
# first layer: underlying points representing individual observations
geom_point(aes(color = species_name), # colored points by species
           shape = 21, # made the points open circle
           alpha = 0.4) + # made points more transparent
# second layer: model prediction line
geom_line(data = preds, # data frame: preds for model 2
          mapping = aes(x = height_cm, # x-axis: height (cm)
                        y = predicted, # y-axis: predicted
                        color = species_name), # colored lines by species
          linewidth = 2) + # increased line width
# third layer: ribbon for 95% confidence interval
geom_ribbon(data = preds, # data frame: preds for model 2
           mapping = aes(x = height_cm, # x-axis: height (cm)
                         y = predicted,
                         ymin = conf.low,
                         ymax = conf.high,
                         fill = species_name), # colored ribbon by species
           alpha = 0.1) + # made ribbon more transparent
# created separate panel for each ant species
facet_wrap(~ species_name,
           # labeled each panel at the top by species name
           labeller = labeller(species_name = c(
             "Crematogaster sjostedti" = "C. sjostedti",
             "Crematogaster mimosae" = "C. mimosae",
             "Crematogaster nigriceps" = "C. nigriceps")))) +
# changed to custom colors for lines and points
scale_color_manual(values = c(
  "Crematogaster sjostedti" = "darkblue",
  "Crematogaster mimosae" = "darkgreen",
```



```

    "Crematogaster nigriceps" = "brown4")) +
# changed to custom colors for ribbons
scale_fill_manual(values = c(
  "Crematogaster sjostedti" = "darkblue",
  "Crematogaster mimosae"   = "darkgreen",
  "Crematogaster nigriceps" = "brown4")) +
# set x-axis range and breaks
scale_x_continuous(limits = c(0, 600),
                   breaks = c(0, 200, 400, 600)) +
# relabeled x- and y-axes
labs (x = "Tree height (cm)",
      y = "Probability of nest occurrence") +
# changed theme from default
theme_minimal() +
# removed panel grids
theme(panel.grid = element_blank()) +
# removed legend
theme(legend.position = "none")

```



i. Write the caption for your figure

Figure 1. Probability of bird nest occurrence increases with tree height across all three acacia ant species. Open circles represent individual tree observations (1 = nest present, 0 = nest absent), colored by ant species. *C. sjostedti* is represented in navy blue, *C. mimosae* in dark green, and *C. nigriceps* in dark brown. Solid lines represent model predicted probability of nest occurrence across the range of observed tree heights (0 to 600 cm), and shaded ribbons represent 95% confidence intervals. Data source: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>.

j. Calculate model predictions

```
# what is the probability of nest occurrence at a value of 600cm (the
# tallest tree in the dataset)?
ggpredict(model2,
           terms = c("height_cm [600]", "species_name"))
```

```
# Predicted probabilities of case_control
```

```
species_name: Crematogaster sjostedti
```

height_cm	Predicted	95% CI
600	0.20	0.00, 0.99

```
species_name: Crematogaster mimosae
```

height_cm	Predicted	95% CI
600	0.74	0.48, 0.90

```
species_name: Crematogaster nigriceps
```

height_cm	Predicted	95% CI
600	1.00	1.00, 1.00

k. Interpret your results

Tree height and ant species interact to predict the probability of bird nest occurrence. As tree height increases, the probability of nest occurrence increases for all three ant species, suggesting that taller trees are generally better nesting sites for birds regardless of ant identity (Figure 1). At 600 cm, the predicted probability of nest occurrence is 0.20 (95% CI [0.00, 0.99]) for *C. sjostedti*, 0.74 (95% CI [0.48, 0.90]) for *C. mimosae*, and 1.00 (95% CI [1.00, 1.00]) for *C. nigriceps*. These differences across ant species likely reflect variation in colony aggression — *C. nigriceps* and *C. mimosae* are known to aggressively defend trees against mammalian herbivores, potentially offering nesting birds greater protection from predators, while *C. sjostedti* provides relatively minimal protection. Ant species identity therefore appears to be an important driver of nesting site presence independent of tree height, consistent with a mutualistic relationship between protective ant colonies and nesting birds.

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

Note: I compared my final affective visualization (presented in Workshop 10; different from the original draft in Homework 3) to the exploratory visualizations from Homework 2.

- **Differences:** My affective visualization displays the logistic regression model (predicted probability of running as a function of sleep duration) within a hand drawn watercolor landscape, using illustrated runners and hourglasses as visual metaphors in place of geometric plot elements. In contrast, the two exploratory visualizations from Homework 2 used standard statistical geometry to prioritize clarity and efficiency over emotional and creative representation. The boxplot with jitter directly compares the distributions of workout duration across three workout types (categorical groups), while the scatterplot shows the relationship between two continuous variables (sleep duration and workout duration) with no model overlaid and no artistic elements.
- **Similarities:** All three visualizations use sleep duration or workout type on the x-axis and show some form of individual observations: the jittered open circles in the boxplot, the individual points in the scatterplot, and the binary observations underlying the logistic curve in the affective visualization.
- **Patterns:** The scatterplot of sleep vs workout duration shows no clear trend; points are scattered without an obvious directional pattern, with most clustering around 60 minutes regardless of sleep duration. This pattern is expected because workout duration includes the lift and yoga workouts, which the boxplot shows are consistently around 60-65 minutes regardless of other factors; meanwhile, runs have a much wider spread (roughly 36-75 minutes) and the lowest median of the three groups. Isolating runs as a

binary outcome in the affective visualization therefore reveals a cleaner pattern — a steep positive relationship between sleep and probability of running — than using duration across all workout types.

- **Feedback:** My original plan was to create my affective visualization from a boxplot with overlaid jitter comparing workout type against sleep duration the previous night. After meeting with An during week 9, I decided to narrow my focus to runs specifically, since runs had the most observations in my dataset and there appeared to be a relationship between sleep duration and whether or not I went on a run. My final affective visualization therefore depicts the probability of going on a run as a function of sleep duration, based on a binomial GLM rather than the boxplot I originally planned.

b. Designing an analysis

I used a binomial GLM to ask whether sleep duration the previous night (continuous, measured in hours) predicts the probability that my workout was a run (binary; 1 = run and 0 = all other workout types). This analysis was appropriate because the response variable is binary, so a standard linear model cannot be used, as the assumption of normally distributed errors would not be met. Therefore, I used a GLM with a binomial error distribution because this test is specifically designed for response variables that have outcomes of 0/1. Sleep is a continuous predictor variable, making it an appropriate continuous explanatory variable in the logistic regression.

c. Check any assumptions and run your analysis

Think about your data

The assumptions for a binomial GLM are met. First, each workout entry represents a separate day, so observations are independent. Second, the response variable is binary, with 1 indicating that a run occurred and 0 indicating that it did not. Finally, the diagnostic plots (simulated below) do not flag any issues with the distribution of residuals or presence of influential outliers.

Write your model

In words: Sleep duration the previous night predicts the occurrence of going on a run (yes = 1, no = 0).

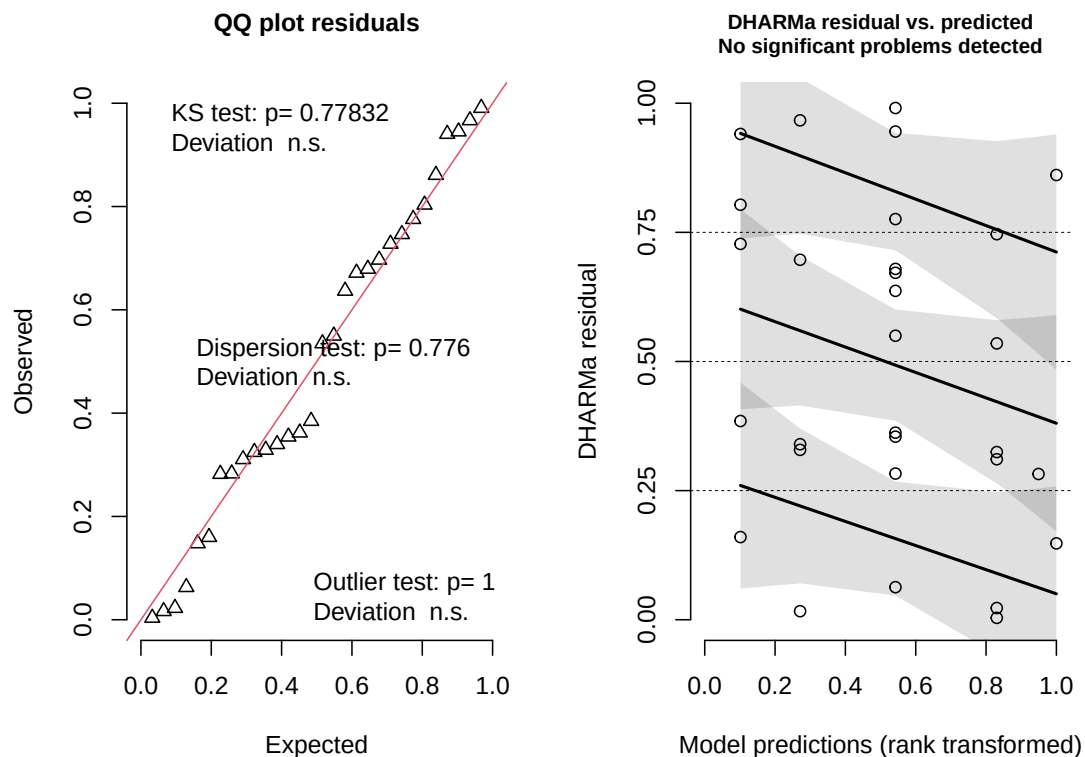
In code:

```
# fitted generalized linear model for running
# response: whether or not workout was a run (binary: 1/0)
# predictor: sleep the previous night (hours)
run_mod <- glm(
  # formula: binary run outcome ~ sleep hours
  is_run ~ sleep_hours,
  # data: my_data data frame
  data = my_data,
  # binomial error distribution for binary response
  family = binomial)
```

Look at the diagnostics

```
# simulated residuals from GLM for diagnostic plots
plot(simulateResiduals(run_mod))
```

DHARMA residual



Information from left plot (QQ plot of residuals):

- **KS test:** Do the residuals follow a uniform distribution? → Generally yes
- **Dispersion test:** Are the residuals more dispersed than expected from a uniform distribution? → No
- **Outlier test:** Are there outliers in the data set that could be influencing model coefficients or predictions? → No

Information from right plot (DHARMA residual vs predicted):

DHARMA diagnostics did not flag any issues, so it is reasonable to say that residuals are randomly arranged across the range of predicted values.

Look at the model coefficients

```
# displayed model summary  
summary(run_mod)
```

Call:

```
glm(formula = is_run ~ sleep_hours, family = binomial, data = my_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-8.7011	3.4964	-2.489	0.0128 *
sleep_hours	1.0873	0.4712	2.307	0.0210 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36.652 on 29 degrees of freedom
Residual deviance: 28.046 on 28 degrees of freedom
AIC: 32.046

Number of Fisher Scoring iterations: 5

```
# calculated inverse logit  
plogis(-8.7011)
```

```
[1] 0.000166375
```

Interpretation:

- inverse logit: 0.00016
- the probability of going on a run if sleep duration is 0 hours is 0.00016

```
# calculated odds ratio
gtsummary::tbl_regression(run_mod,
                           exponentiate = TRUE) |>
# converted to gt object for formatting
as_gt() |>
# rendered table to correct position
tab_options(latex.tbl.pos = "h",          # h for render the table "here"
             latex.use_longtable = TRUE,  # set PDF rendering option
             table.width = px(500))      # made the table wider
```

Characteristic	OR	95% CI	p-value
sleep_hours	2.97	1.37, 9.40	0.021

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Interpretation:

- odds ratio: 2.97
- $2.97 > 1$ so the probability of going on a run increases by a factor of 2.97 with every 1 hour increase in sleep (95% CI: [1.37, 9.40])

Look at the model predictions

```
# what is the probability of going on a run after sleeping for 10 hours?
ggpredict(run_mod,
           terms = "sleep_hours [10]")
```

Predicted probabilities of is_run

sleep_hours	Predicted	95% CI
10	0.90	0.39, 0.99

```
# what is the probability of going on a run after sleeping for 4 hours?
ggpredict(run_mod,
          terms = "sleep_hours [4]")
```

Predicted probabilities of is_run

sleep_hours	Predicted	95% CI
4	0.01	0.00, 0.25

```
# what is the probability of going on a run after sleeping for 7 hours
# (the average sleep duration in my dataset)?
ggpredict(run_mod,
          terms = "sleep_hours [7]")
```

Predicted probabilities of is_run

sleep_hours	Predicted	95% CI
7	0.25	0.11, 0.47

Note:

- If I slept for 10 hours, the probability of going on a run the next day is 0.90 (95% CI [0.39, 0.99]).
- If I slept for 4 hours, the probability of going on a run the next day is 0.01 (95% CI [0.00, 0.25]).
- If I slept for 7 hours, the probability of going on a run the next day is 0.25 (95% CI [0.11, 0.47]).

d. Create a visualization

Visualize model predictions

```
# generated model predictions across range of sleep values in data set
mod_preds <- ggpredict(run_mod,
                      terms = "sleep_hours [5:10 by = 1]")

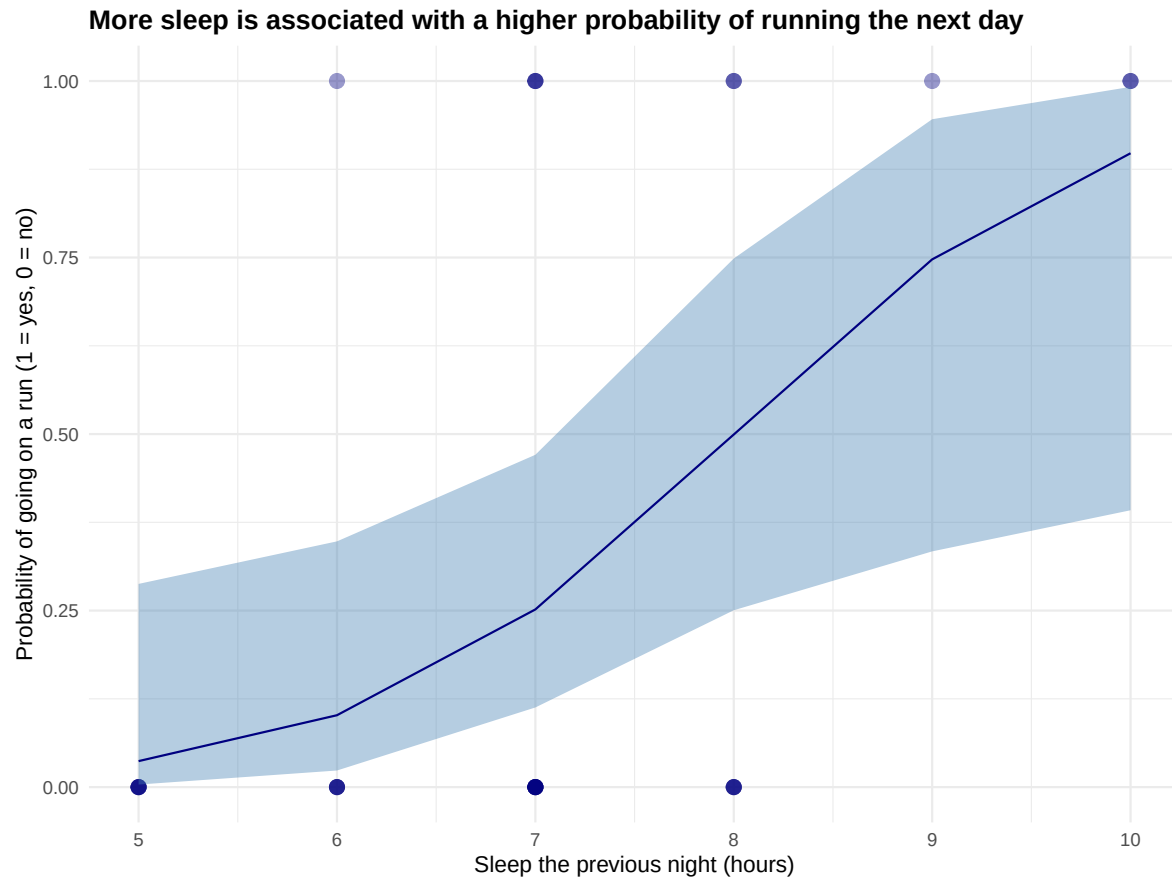
# base layer: ggplot with raw data
```



```

ggplot(my_data,          # data frame: my_data
       aes(x = sleep_hours, # x-axis: sleep duration
           y = is_run)) + # y-axis: probability of running (0/1)
# first layer: raw observations
geom_point(size = 3,          # made points larger
            alpha = 0.4,      # made points more transparent
            color = "navyblue") + # changed points to navy blue
# second layer: 95% confidence interval ribbon from model predictions
geom_ribbon(data = mod_preds, # data frame: model predictions
            aes(x = x,
                y = predicted,
                ymin = conf.low,
                ymax = conf.high),
            fill = "steelblue", # changed ribbon color to steel blue
            alpha = 0.4) +      # made ribbon more transparent
# third layer: logistic curve from model predictions
geom_line(data = mod_preds,
           aes(x = x,
               y = predicted),
           color = "navyblue") + # changed line color to navy
# added title
# relabeled x- and y-axis
labs(title =
"More sleep is associated with a higher probability of running the next day",
      x = "Sleep the previous night (hours)",
      y = "Probability of going on a run (1 = yes, 0 = no)") +
# changed theme from default
theme_minimal() +
# made title bold
theme(plot.title = element_text(face = "bold")) +
# removed legend
theme(legend.position = "none")

```



Describe the model

```
# displaying model in table
as_flextable(run_mod)
```

	Estimate	Standard Error	z value	Pr(> z)	
(Intercept)	-8.701	3.496	-2.489	0.0128	*
sleep_hours	1.087	0.471	2.307	0.0210	*

Estimate	Standard Error	z value	Pr(> z)
<i>Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '.' < 0.05</i>			

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36.65 on 29 degrees of freedom

Residual deviance: 28.05 on 28 degrees of freedom

e. Write a caption

Figure 2. Longer sleep duration is associated with a higher probability of going on a run the next day. Navy circles represent individual workout observations (0 = not a run, 1 = run). The navy blue line represents model predicted probability of running across the range of observed sleep values (5-10 hours), and the shaded blue ribbon represents the 95% confidence interval around model predictions. Data collected from personal fitness tracking records between 2026-04-14 and 2026-05-26.

f. Write about your results

I tend to go on runs more often on days following nights with more sleep. For every 1 hour increase in sleep the previous night, the odds of going on a run increased by a factor of 2.97 (95% CI: [1.37, 9.40], $p = 0.021$, $\alpha = 0.05$). After sleeping for 10 hours, the probability of going on a run the next day is 0.90 (95% CI [0.39, 0.99]), compared to a probability of 0.01 (95% CI [0.00, 0.25]) after sleeping for only 4 hours. These results suggest that sleep duration substantially influences the probability of running the following day.

g. Sharing your affective visualization

Presented visualization in Week 10 Workshop.